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The Effect of the Angle of Rocket Fins on the Maximum Vertical Flight Height of Rockets

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Introduction:

For millions of people across the globe, model rockets are an exciting hobby. If you go to a local park, you can often see these people launching their rockets into the atmosphere. Even though it appears to just be a simple hobby on the surface, it's all about precision. Small adjustments to the rocket or the launch pad have drastic effects on flight behavior (*Basic Rocketry Aerodynamics*). This experiment will be testing to see how the angle of the rocket fin affects the vertical flight height, which is also known as the apex of the flight. The fins themselves are a key component of the rocket. They provide both stability and the ability to alter the flight path, whether it is beneficial, harmful or both (Nakka, *Richard Nakka's Experimental Rocketry Site*). The research examines the impact changing the angle of the fins in relation to the body of the rocket on maximum vertical flight height. In order to ensure consistent results, the two rockets will have to have congruent specifications that optimize their performance (*The Aerodynamics of Model Rockets Part 3- Rocket Design Analysis*). A full list of constants can be found in Materials and Methods: Section B. However, not everything can be kept constant so several assumptions must be made which include the margin of error on the altimeters and having the same amount of thrust produced by the rocket engines at every launch.

It has already been shown that changing various attributes of the rocket fins produced a broad spectrum of flight behavior (Kim, Lee and Tahk, *New Structure For An Aerodynamic Fin Control System For Tail Fin-Controlled STT Missiles*). This experiment will serve to clarify and further the results of flight behavior (in this case, peak vertical height in meters) in respect to different fin angles in relation to the body of the rocket. The results of this project project could benefit more than just model rocket enthusiasts. Fins used for guiding projectiles can be found in a variety of other disciplines, such as military sciences where fins on artillery shells are used to

direct their path (Garner, Weinacht and Kaste, *Experimental Validation Of Elliptical Fin-Opening Behavior*). As such, the results from this project would potentially allow for a more efficient use of the thrust produced in projectile propulsion.

Hypothesis:

Alternate Hypothesis (H_1): *Rocket fins placed at a 45° angle to the body will result in a significantly higher peak vertical flight height than the peak vertical flight height of fins placed at 90° angle to the body.*

Null Hypothesis (H_0): Rocket fins placed at a 45° angle to the body will not result in a significantly higher peak vertical flight height than the peak vertical flight height of fins placed at 90° angle to the body.

Materials and Methods:

A. Experimental Procedure

1) Build model rockets according to directions from package (do not attach fins yet)
2) Attach fins at predetermined angle in relation to rocket body (90° fins)
3) Calibrate altimeter according to the instructions that came with it
4) Attach altimeter to rocket
5) Travel to a suitable launch area (like an open field)- have an adult present for safety reasons
6) Check for favorable weather conditions (no precipitation, still air) and that all constants are kept consistent
7) Place rocket engine in engine chamber
8) Place rocket on launch pad (be sure to follow any directions that the launch pad contains)
9) Attach engine igniter and igniter plug
10) Activate engine igniter
11) Launch rocket
12) Retrieve rocket
13) Check altimeter for reading
14) Record the peak vertical height data from the altimeter in raw data tables
15) Repeat steps 6-15 eleven more times
16) Repeat steps 2-15 twelve times using 45 degree fins instead of 90 degree fins on second rocket

17) Analyze data (see section Data Collection)
18) Make conclusions from the aforementioned data analysis

B. Experimental Variables and Controls:

Independent variable: Angle of Rocket fin in relation to the rocket body

Dependent variable: Maximum vertical flight height (meters)

Levels:	90° fin angle	45° fin angle
# trials:	12 launches	12 launches
Control?	control group	test group

Constants

Direction of travel	Weather (no precipitation)
Wind speed (still air)	Type of rocket
Launch pad (same one for all launches)	Type of rocket engine
Temperature (<90°F and >40°F)	Material of rocket fins
Location	Location of rocket fins on the rocket

C. Data Collection:

Data Table of Apex Height for Each Rocket Fin Angle (meters)

Launch Number	90° fin angle (Control)	45° fin angle (Test Group)
1		
12		

Data Analysis:

In order to conclude whether or not the 45 degree fins produce a significantly higher peak vertical flight height than the peak vertical flight height caused by 90 degree fins, a one tailed t-test will be used to obtain a p-value. This p-value will then be referenced to an alpha of 0.05.

D. Safety Consideration:

Risks-

Hazardous chemicals, activities and devices:

- model rocket engines (size A8-3)
- model rocket igniters
- launching device/pad
- model rockets

Risks:

- engines could explode
- electrocution by launch device
- igniters could explode
- rockets could hit someone

Safety Precautions:

- Eye protection will be worn at all times
- Launches will be done on designated area
- Appropriate launch and set up procedures will be followed
- An adult will supervise at all times
- Used rocket engines will be soaked in water for 24 hours to ensure unreactivity of the engines

Budget:

Quantity	Item	Cost per unit	Notes	Total Cost
2	Estes 2178 Hi-Flier Flying Model Rocket Kit	$8.23	-	$16.46
9	Estes A8-3 Engine Pack (3-Each)	$8.95	-	$80.55
1	Estes Cox 1469 Tandem X Launch Set	$23.35	-	$23.35
1	Jolly Logic AltimeterOne	$49.95	$49.95-Research group	$49.95
5	Estes 302301 Model Rocket Igniters	$7.95	-	$39.75
Total Cost				$210.06
Total Request from NJAS				$150

Bibliography

"Basic Rocketry Aerodynamics"*Dark.DK Technotes*. Dark.DK. Web. 07 May 2014 http://www.dark.dk/archive/technotes/BaseAero.pdf

This article is relevant to my project since it provides a basic explanation of model rockets as a whole. In addition, this article details how factors outside of the actual rocket, such as launch angle, can cause a variation in performance. If I wish to be as accurate as possible, these factors must be kept the same.

Garner, James M., Paul Weinacht, and Robert P. Kaste. "Experimental Validation Of Elliptical Fin-Opening Behavior." Shock & Vibration 10.2 (2003): 115. Academic Search Premier. Web. 7 May 2014.

This article relates to my project since it discusses how the deployment of fins on an artillery projectile at various angles while in motion can change the overall precision in launching aforementioned projectile. This is exceedingly relevant to my project since I will be preforming a similar experiment, although my fins will be stationary and not deployed mid-flight.

Kim, Seung-Hwan, Yong-In Lee, and Min-Jea Tahk. "New Structure For An Aerodynamic Fin Control System For Tail Fin-Controlled STT Missiles." Journal Of Aerospace Engineering 24.4 (2011): 505-510. Academic Search Premier. Web. 7 May 2014.

This article is relevant to my experiment since it discusses the effect of fins on a rocket in flight. The article discusses how with a slight change in the angle or position of a fin, the rocket's flight path can change drastically. Since I will be changing the angle of the fins on my rocket, this article provides some basic information of what to possibly expect.

Nakka, Richard. "Richard Nakka's Experimental Rocketry Site." Richard Nakka's Experimental Rocketry Site. Nakka Rocketry, 26 Aug. 2001. Web. 02 May 2014. http://www.nakka-rocketry.net/fins.html

This article is relevant to my experiment since it details the purpose of fins. The article also the various forces that act upon the rocket fins and how to counteract forces that negatively influence the performance of the rocket. Within this article there are also instructions that detail how to stabilize rockets and therefore more efficiently channel the trust produced by the engine.

"The Aerodynamics of Model Rockets Part 3- Rocket Design Analysis | Articles from Sigma Rockets." Articles from Sigma Rockets RSS. Sigma Rockets, 29 Apr. 2012. Web. 04 May 2014. http://www.sigmarockets.com/blog/2012/04/the-aerodynamics-of-model-rockets-part-3-rocket-design-analysis/

This website relates to my experiment since the article explains the parts of a model rocket and their influence on the overall aerodynamics and behavior. The article details the optimal lengths and shapes for the various model rocket components in order to achieve maximum performance.